Quantum Secret Sharing and Implications for Financial Infrastructure

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1 Introduction to Secret Sharing and the Quantum Threat

Modern financial systems rely on cryptographic keys to authorize transactions, authenticate institutions, and secure communication. Certain keys, particularly long-term, high-value keys, play a foundational role: they initialize secure systems, authorize configuration changes, and control access to other keys. Compromise of such a key does not result in localized failure, but in the potential loss of control over an entire system.

A natural approach is to attempt to secure these keys as completely as possible: to store them in hardened devices, restrict access to trusted administrators, and enforce strict operational controls. However, this approach leaves a structural vulnerability. If any single entity, whether a person, device, or process, retains full control over the key, then compromising that entity is sufficient to compromise the key.

In practice, this risk is addressed by eliminating single points of control. Rather than storing a key in one place, control over it is distributed across multiple parties such that no individual can use or reconstruct it alone.

The Society for Worldwide Interbank Financial Telecommunication (SWIFT), which connects over 11,000 financial institutions across more than 200 countries and coordinates trillions of dollars in daily transfers [21], provides a representative example. In SWIFT deployments, high-value cryptographic keys—such as those used to initialize hardware security modules (HSMs), authorize configuration changes, or enable the use of other keys—are not held by any single administrator. Instead, they are divided into multiple components, each assigned to a different custodian [20].

This division is enforced during specific, high-risk operations. When an HSM is initialized, when a master key is activated, or when a system must be recovered after failure, a predefined subset of custodians must be physically present and must provide their respective components. These components are combined within the secure boundary of the HSM to enable the operation. No single custodian can act independently, and the key is never exposed in full outside the device.

This mechanism does not apply to routine transaction processing. Financial institutions continue to authorize and transmit messages independently; there is no network-wide consensus or collective approval process. The distribution applies only to the control of high-value keys. Its purpose is specific: to ensure that no single insider, device, or point of failure can unilaterally compromise them.

This operational pattern raises a precise question: how can a secret be divided among multiple parties such that only authorized groups can reconstruct or use it, while smaller groups cannot?

1.1 The Secret Sharing Problem

Abstracted from the operational setting, the requirement can be stated formally. A secret must be distributed among n parties such that any sufficiently large subset of parties can reconstruct it, and any smaller subset obtains no information about it and cannot act on it.

This is the secret sharing problem, independently formalized by Shamir [17] and Blakley [4] in 1979. A scheme satisfying this property with threshold k is called a (k, n)-threshold scheme: within the n secret keepers, any group of k parties can reconstruct the secret, while any group of fewer than k parties learns nothing. Among the proposed constructions, Shamir's scheme is the most widely deployed due to its simplicity and strong security properties; Blakley's independent construction, based on hyperplane geometry, establishes the same threshold guarantee by a different algebraic route but is less commonly used in practice. For our purposes, we only discuss the Shamir's scheme.

1.2 Shamir's Secret Sharing: Construction

Shamir's construction defines a (k, n)-threshold scheme over a finite field $\mathbb{F}_p$, with the secret $s \in \mathbb{F}_p$:

- p: a large prime defining the finite field $\mathbb{F}_p$ in which all arithmetic is performed,
- n: the number of parties (and therefore the number of shares generated),
- k: the reconstruction threshold, with $1 \leq k \leq n$.

Construction. Shamir's scheme relies on polynomial interpolation over $\mathbb{F}_p$. The dealer constructs a random polynomial of degree $k-1$:

$$f(x) = s + a_1 x + a_2 x^2 + \cdots + a_{k-1} x^{k-1} \pmod{p}, \tag{1}$$

where coefficients $a_1, \ldots, a_{k-1} \in \mathbb{F}_p$ are chosen uniformly at random.

The secret is embedded as the constant term:

$$f(0) = s. \tag{2}$$

Share generation. The scheme generates n shares by evaluating the polynomial at n distinct, publicly known, nonzero points in $\mathbb{F}_p$, typically

$$x_i \in \{1, 2, \ldots, n\}.$$

Each party N_i receives a share

$$s_i = (x_i, f(x_i)). \tag{3}$$

Threshold structure. The construction enforces two regimes:

- Any k shares provide k independent constraints on a degree $k-1$ polynomial, uniquely determining all coefficients, including s.
- Any $k-1$ shares leave at least one degree of freedom, so multiple polynomials remain consistent with the observed data, leaving s undetermined.

Reconstruction. Given k shares, the polynomial is recovered via Lagrange interpolation, and the secret is obtained as:

$$s = f(0) = \sum_{i=1}^{k} y_i \prod_{\substack{j=1 \\ j \neq i}}^{k} \frac{-x_j}{x_i - x_j} \pmod{p}. \tag{4}$$

A very simple example. Let the secret be $s = 7$, distributed among 3 parties $n = 3$. Two parties are enough to reconstruct $k = 2$. Random prime number is $p = 11$. Choosing first coefficient $a_1 = 3$ gives:

$$f(x) = 7 + 3x \pmod{11}, \tag{5}$$

What each party receives:

$$N_1 = (1,\ 10), \qquad N_2 = (2,\ 2), \qquad N_3 = (3,\ 5). \tag{6}$$

Each share corresponds to a point on the line defined by $f(x)$. Two shares suffice because two points uniquely determine a line. Using $s_1 = (1, 10)$ and $s_2 = (2, 2)$, the slope is recovered as:

$$\frac{2 - 10}{2 - 1} = -8 \equiv 3 \pmod{11}, \tag{7}$$

which recovers the coefficient $a_1 = 3$. Each party can then work out the constant term $s = 7$.

A single share, however, leaves one degree of freedom. The point $(1, 10)$ is consistent with many different lines over $\mathbb{F}_{11}$, each corresponding to a different secret. For any $s' \in \mathbb{F}_{11}$, the polynomial:

$$f'(x) = s' + (10 - s')x \pmod{11} \tag{8}$$

passes through $(1, 10)$. Each such polynomial yields a different value at $x = 0$, yet produces the same observed share. A single share therefore does not reduce uncertainty about the secret: every value in $\mathbb{F}_{11}$ remains equally possible. Only when enough shares are combined to eliminate all degrees of freedom does the secret become determined.

1.3 Security Guarantee and Scope

Shamir's scheme provides an information-theoretic security guarantee below the threshold [17]. Any adversary holding fewer than k shares gains zero information about the secret, regardless of computational resources. This guarantee does not rely on assumptions about the difficulty of any mathematical problem, which explains why threshold schemes based on Shamir's construction are widely used for protecting high-value cryptographic keys.

However, this guarantee is limited in scope. It applies only to the shares as mathematical objects and assumes a passive adversary who collects shares and attempts algebraic reconstruction. In practice, the scheme is embedded within a broader cryptographic environment: shares must be transmitted over secure channels, custodians must be authenticated before participating in reconstruction, and the reconstructed secret is used within other cryptographic protocols. These surrounding components rely on standard primitives such as RSA, elliptic-curve Diffie–Hellman (ECDH), and digital signature schemes. The security of the overall system therefore depends on these mechanisms in addition to the threshold structure itself.

1.4 The Quantum Vulnerability

This dependence becomes a point of failure in the presence of a quantum adversary. Shor demonstrated that both integer factorization and the discrete logarithm problem can be solved in $\mathcal{O}\big((\log N)^3\big)$ time on a sufficiently powerful quantum computer, using quantum phase estimation and the quantum Fourier transform [18, 19]. Since RSA, ECDSA, and ECDH each reduce to one of these problems, their security would not survive such a development.

The implication for threshold schemes is indirect but significant. Shamir's scheme itself remains secure against a computationally unbounded classical adversary—its information-theoretic guarantee is unchanged. However, the infrastructure used to distribute shares and authenticate participants does not share this immunity. An adversary need not attack the threshold structure directly; compromising the surrounding classical infrastructure is sufficient.

This risk has already prompted institutional responses. In 2024, NIST finalized its first post-quantum cryptographic standards, selecting CRYSTALS-Kyber (FIPS 203) for key encapsulation and CRYSTALS-Dilithium (FIPS 204), FALCON (FIPS 206), and SPHINCS+ (FIPS 205) for digital signatures [12, 13, 14]. Financial regulators have begun requiring institutions to inventory their cryptographic exposure accordingly [2].

Post-quantum cryptography addresses the problem by adopting alternative hardness assumptions—such as those based on lattices, error-correcting codes, or hash functions—for which no efficient quantum algorithms are currently known [3]. These standards represent the near-term migration path for most institutions. However, they remain computational guarantees: their security depends on assumptions about what cannot be efficiently computed, assumptions that are unproven and in principle subject to future revision.

1.5 From Computational to Physical Security

Quantum Secret Sharing (QSS) takes a fundamentally different approach. Rather than relying on computational assumptions, it derives security from the physical properties of quantum systems.

The key principle is that quantum information cannot be observed without disturbance [15]. In protocols based on multipartite entanglement, the information encoding the secret is distributed across a joint quantum state shared among multiple parties. Any attempt by an unauthorized subset of parties to extract information necessarily disturbs this state in a detectable way.

The foundational result, due to Hillery, Bužek, and Berthiaume [10], establishes that such schemes are possible: authorized subsets can reconstruct the secret, while unauthorized subsets cannot gain information without revealing their presence. The distinction from classical secret sharing is fundamental. Classical schemes are secure because certain computations are believed to be infeasible. Quantum schemes are secure because certain operations are physically constrained.

The following sections develop this framework in detail. Section II introduces the multipartite entanglement structures that make QSS possible. Section III presents the HBB protocol concretely. Section IV formalizes the security guarantees. Sections V discuss generalizations and the practical path to deployment.

2 Multipartite Entanglement and GHZ States

The security of classical secret sharing, as established in last section, ultimately rests on computational assumptions. The quantum approach replaces these with a physical guarantee: information about the secret is encoded not in any local system, but in the correlations of a shared quantum state. No unauthorized subset of parties can access these correlations without disturbing the state in a detectable way. Realizing this requires an entanglement structure that is multipartite—one that cannot be reduced to pairwise correlations between subsystems.

2.1 From Bipartite to Multipartite Entanglement

Two qubits are said to be entangled if their joint state cannot be written as a product of individual qubit states. The canonical example is the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big). \tag{9}$$

Measuring either qubit in the computational basis yields a uniformly random outcome, but the two outcomes are perfectly correlated: if one party observes 0, the other is guaranteed to observe 0 as well, and similarly for 1. This nonclassical correlation is the resource exploited by two-party quantum key distribution protocols such as E91 [7].

Secret sharing, however, is multipartite. The two-qubit entanglement is insufficient for this purpose. Instead, the Greenberger–Horne–Zeilinger (GHZ) state can be used for this purpose [9]. Their analysis showed that the three-qubit quantum correlations are inconsistent with local hidden variable theories not merely statistically, as with Bell's inequality, but as a matter of deterministic prediction on single experimental runs. It is this stronger nonclassicality that makes GHZ states the natural entanglement resource for quantum secret sharing.

2.2 The Three-Qubit GHZ State

The three-qubit GHZ state is defined as:

$$|\mathrm{GHZ}\rangle = \frac{1}{\sqrt{2}}\big(|000\rangle + |111\rangle\big). \tag{10}$$

Three structural properties of this state are relevant to secret sharing.

Maximal bipartite entanglement across every cut. Tracing out any single qubit leaves the remaining two in a maximally mixed state. For example, tracing out qubit C:

$$\rho_{AB} = \mathrm{Tr}_C\big(|\mathrm{GHZ}\rangle\langle\mathrm{GHZ}|\big) = \frac{1}{2}\big(|00\rangle\langle 00| + |11\rangle\langle 11|\big). \tag{11}$$

This is not an entangled state. Parties A and B share no entanglement with each other once C is discarded. The same holds for every bipartition. No two-party subset retains entanglement; only classical correlations remain.

Perfect correlation in the computational basis. Measuring any one qubit in the $\{|0\rangle, |1\rangle\}$ basis collapses the state to either $|000\rangle$ or $|111\rangle$ with equal probability. The measuring party obtains a uniformly random outcome and learns nothing about the global state. The remaining two qubits are left in a product state $|aa\rangle$ for $a \in \{0, 1\}$, determined by the measurement outcome but unknown to any party who has not communicated with the measurer.

Permutation symmetry. The state (10) is invariant under any permutation of its three qubits. No party occupies a distinguished role in the entanglement structure.

Together, these properties establish the central structural feature: the measurement outcomes of any two parties, taken in isolation, are uniformly random and mutually independent. No two-party subset can determine anything about the third party's outcome. Yet the outcomes of all three parties together satisfy a deterministic constraint, enabling the full group to reconstruct a shared value that no strict subset could have computed. This is precisely the correlation structure required for a $(2,3)$-threshold secret sharing scheme.

2.3 Preparation of the GHZ State

The GHZ state can be prepared from the computational basis state $|000\rangle$ using two standard quantum gates: the Hadamard gate H and the controlled-NOT gate CNOT. The preparation proceeds in three steps.

Step 1. Apply H to the first qubit:

$$H|0\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big) \quad \Longrightarrow \quad \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big)|00\rangle. \tag{12}$$

Step 2. Apply $\mathrm{CNOT}_{1\to 2}$, with qubit 1 as control and qubit 2 as target:

$$\frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big)|00\rangle \quad \Longrightarrow \quad \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big)|0\rangle. \tag{13}$$

Step 3. Apply $\mathrm{CNOT}_{1\to 3}$, with qubit 1 as control and qubit 3 as target:

$$\frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big)|0\rangle \quad \Longrightarrow \quad \frac{1}{\sqrt{2}}\big(|000\rangle + |111\rangle\big) = |\mathrm{GHZ}\rangle. \tag{14}$$

This three-gate circuit is the canonical preparation protocol and is directly implementable on any platform supporting H and CNOT gates, including current superconducting qubit architectures.

2.4 Measurement Statistics and the Secret Sharing Structure

The properties established in Section 2.2 describe the GHZ state in the computational basis. To understand why it supports secret sharing, it is necessary to examine its behavior under measurements in the X and Y bases, since it is these measurements that the HBB protocol employs.

Define the eigenstates of the Pauli X and Y operators as:

$$|\pm\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle \pm |1\rangle\big), \tag{15}$$

$$|\pm i\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle \pm i\,|1\rangle\big). \tag{16}$$

Expressing $|0\rangle$ and $|1\rangle$ in terms of these eigenstates and substituting into (10), the GHZ state can be rewritten in the X basis as:

$$|\mathrm{GHZ}\rangle = \frac{1}{2}\big(|{+}{+}{+}\rangle + |{+}{-}{-}\rangle + |{-}{+}{-}\rangle + |{-}{-}{+}\rangle\big). \tag{17}$$

This expansion reveals the correlation structure directly. If all three parties measure in the X basis with outcomes $x_A, x_B, x_C \in \{+1, -1\}$, only the four terms in (17) contribute. Each term contains an even number of $|-\rangle$ states, so the product of outcomes satisfies:

$$x_A \cdot x_B \cdot x_C = +1 \quad \text{with certainty.} \tag{18}$$

A similar analysis applies when one party measures in the Y basis and the remaining two measure in X. Rewriting $|\text{GHZ}\rangle$ in the mixed YXX basis:

$$|\text{GHZ}\rangle = \frac{1}{2}\big(|{+i},+,+\rangle - |{+i},-,-\rangle - |{-i},+,-\rangle - |{-i},-,+\rangle\big), \tag{19}$$

from which it follows that:

$$y_A \cdot x_B \cdot x_C = -1 \quad \text{with certainty,} \tag{20}$$

and by symmetry, $x_A \cdot y_B \cdot x_C = -1$ and $x_A \cdot x_B \cdot y_C = -1$ as well. When all three parties measure in the Y basis:

$$y_A \cdot y_B \cdot y_C = +1 \quad \text{with certainty.} \tag{21}$$

These constraints can be summarized compactly. Let each party choose independently between the X and Y basis, and let n_Y denote the number of parties who chose Y. Then:

$$\prod_{i \in \{A,B,C\}} r_i = (-1)^{\frac{n_Y(n_Y-1)}{2}} \tag{22}$$

Crucially, while the product of all three outcomes is deterministic, the marginal distribution of any two outcomes is uniform. Given only r_A and r_B, for instance, r_C is uniformly distributed over $\{+1, -1\}$ conditioned on any fixed values of r_A and r_B—the two observed outcomes impose no constraint on the third. This is the measurement-level manifestation of the structural property identified in Section 2.2: information is encoded globally, in the three-party correlation, and not in any local or pairwise subsystem.

2.5 Why Classical Shared Randomness Cannot Reproduce These Correlations

The correlation structure described above might appear, at first glance, to be reproducible by classical means. Suppose three parties share a hidden variable λ drawn from some distribution, and each party's measurement outcome is a deterministic function of λ and their chosen basis. Could such a model reproduce the constraints (18)–(21)?

Greenberger, Horne, Shimony, and Zeilinger showed in 1990 that it cannot [8]. The argument is direct. Assume a local hidden variable model assigns definite values $x_A, x_B, x_C \in \{+1, -1\}$ and $y_A, y_B, y_C \in \{+1, -1\}$ to the outcomes of X and Y measurements for each party. The constraints from (18) and (20) and its permutations require:

$$x_A \cdot x_B \cdot x_C = +1, \tag{23}$$
$$y_A \cdot x_B \cdot x_C = -1, \tag{24}$$
$$x_A \cdot y_B \cdot x_C = -1, \tag{25}$$
$$x_A \cdot x_B \cdot y_C = -1. \tag{26}$$

Multiplying (24), (25), and (26) together:

$$(y_A \cdot x_B \cdot x_C)(x_A \cdot y_B \cdot x_C)(x_A \cdot x_B \cdot y_C) = x_A^2 \cdot x_B^2 \cdot x_C^2 \cdot y_A \cdot y_B \cdot y_C = y_A \cdot y_B \cdot y_C = (-1)^3 = -1. \tag{27}$$

Here we used $x_i^2 = 1$ for all i. But the quantum mechanical prediction (21) requires $y_A \cdot y_B \cdot y_C = +1$. The two results are contradictory. No assignment of definite local values can simultaneously satisfy all four constraints, and therefore no local hidden variable model can reproduce the GHZ correlations [8, 11].

This result has a direct implication for security. The correlations that the HBB protocol exploits are not merely statistically nonclassical—they are logically impossible to simulate with any classical shared resource. An adversary attempting to replace the quantum channel with a classical simulation cannot reproduce the required correlation structure, and any such attempt will be detectable. The security of the protocol therefore rests not on a computational barrier but on a logical one.mu

3 The HBB Quantum Secret Sharing Protocol

This section shows how the properties of GHZ state are turned into a working secret sharing protocol. The scheme, developed by Hillery, Bužek, and Berthiaume [10], was the first complete quantum secret sharing protocol and remains the foundation of the field.

3.1 Setting and Goal

The protocol involves three parties: a dealer Alice (A) and two agents Bob (B) and Charlie (C). Alice holds a classical secret bit $s \in \{0, 1\}$, equivalent to one bit of a cryptographic key in practice, and wishes to distribute it such that Bob and Charlie together can recover s, but no one alone can learn anything about it. This is like a $(2, 3)$-threshold scheme.

Alice, Bob, and Charlie each hold one qubit of a shared GHZ state, prepared and distributed before the protocol begins. Classical communication channels between all parties are authenticated but not assumed to be private. The quantum channel used to distribute the GHZ state is assumed to be noiseless for the purpose of this presentation.

3.2 Protocol Description

Step 1: GHZ State Distribution Alice prepares the three-qubit GHZ state:

$$|\mathrm{GHZ}\rangle = \frac{1}{\sqrt{2}}\big(|000\rangle + |111\rangle\big), \tag{28}$$

and distributes qubit B to Bob and qubit C to Charlie, retaining qubit A.

Step 2: Alice and Bob Measure their qubit Alice measures her qubit in either the X or Y basis, chosen at random. She records her basis choice $b_A \in \{X, Y\}$ and her outcome $r_A \in \{+1, -1\}$. Similarly, Bob independently measures his qubit in either the X or Y basis. He records his basis choice b_B and his outcome r_B.

Step 3: Basis Announcement Alice and Bob each announce their basis choices b_A and b_B publicly over the classical channel. Charlie, who has not yet measured, now knows both basis choices.

Step 4: Charlie's Measurement Charlie measures his qubit in a basis b_C determined by b_A and b_B according to the GHZ correlation constraints derived in Section 2. Specifically, Charlie selects his basis according to Table 1. Charlie records his outcome $r_C \in \{+1, -1\}$.

b_A	b_B	b_C
X	X	X
X	Y	Y
Y	X	Y
Y	Y	X

Table 1: Charlie's measurement basis as a function of Alice's and Bob's basis choices. The selection rule ensures that the triple (b_A, b_B, b_C) always falls into one of the four correlation classes established in Section 2.

Step 5: Secret Encoding Alice encodes her secret bit s by deciding how to announce her measurement result. She computes a modified result $\tilde{r}_A$ as:

$$\tilde{r}_A = (-1)^s \cdot r_A, \tag{29}$$

and announces $\tilde{r}_A$ publicly. If $s = 0$, Alice announces her true outcome. If $s = 1$, she announces the flipped outcome. This single classical bit is the only information about s that enters the classical channel.

Step 7: Reconstruction Bob and Charlie cooperate to recover s. From the GHZ correlation constraints of Section 2, the three measurement outcomes always satisfy:

$$r_A \cdot r_B \cdot r_C = c(b_A, b_B, b_C), \tag{30}$$

where $c(b_A, b_B, b_C) \in \{+1, -1\}$ is the deterministic constant determined by the basis combination.

Substituting (29) into (30) and solving for s:

$$\begin{aligned} \tilde{r}_A \cdot r_B \cdot r_C &= (-1)^s \cdot r_A \cdot r_B \cdot r_C \\ &= (-1)^s \cdot c(b_A, b_B, b_C). \end{aligned} \tag{31}$$

Since Bob and Charlie know $\tilde{r}_A$ (announced by Alice), r_B, r_C, and $c(b_A, b_B, b_C)$ (determined by the public basis announcements), they can solve for s as:

$$(-1)^s = \frac{\tilde{r}_A \cdot r_B \cdot r_C}{c(b_A, b_B, b_C)}, \tag{32}$$

from which s is recovered uniquely. Neither Bob nor Charlie can perform this computation alone, since Bob does not know r_C and Charlie does not know r_B until they communicate.

3.3 Experiment on IBM Q

The four measurement circuits described above were implemented and executed on the IBM Quantum Composer statevector simulator with 1024 shots each. Each circuit prepares the GHZ state and applies the appropriate basis-change gates for one of the four valid (b_A, b_B, b_C) combinations, with $s = 0$ in all cases. The statevector simulator provides a noiseless execution environment, allowing the protocol's logical correctness to be verified independently of hardware-level gate errors and decoherence.

3.3.1 Circuit Construction

All four circuits share a common GHZ preparation stage: a Hadamard gate on qubit $q[0]$ (Alice) followed by CNOT gates from $q[0]$ to $q[1]$ (Bob) and from $q[0]$ to $q[2]$ (Charlie). Measurement in the X basis is implemented by applying a Hadamard gate before measurement; measurement in the Y basis is implemented by applying $S^\dagger$ followed by H before measurement, where $S^\dagger$ is the inverse phase gate. The circuits are summarized in Table 2.

Circuit	b_A	b_B	b_C	s
1	X	X	X	0
2	X	Y	Y	0
3	Y	X	Y	0
4	Y	Y	X	0

Table 2: The four circuits executed on the IBM Quantum Composer simulator. All circuits encode $s = 0$; the remaining four circuits with $s = 1$ are symmetric and were verified separately.

3.3.2 Results

The measurement outcomes for all four circuits are shown in Figure 1. In each case, probability is concentrated on exactly the four bitstrings consistent with the GHZ correlation constraint for that basis combination. Defining the reconstruction fidelity as the fraction of shots whose outcomes satisfy the deterministic product constraint, the results are summarized in Table 3.

Circuit	b_A	b_B	b_C	$c(b_A, b_B, b_C)$	Fidelity
1	X	X	X	$+1$	97.3%
2	X	Y	Y	-1	92.0%
3	Y	X	Y	-1	91.4%
4	Y	Y	X	-1	95.4%
Mean fidelity					94.0%

Table 3: Reconstruction fidelity for each circuit, defined as the fraction of 1024 shots satisfying the deterministic product constraint $(-1)^{c[0]}\cdot(-1)^{c[1]}\cdot(-1)^{c[2]} = (-1)^s\cdot c(b_A, b_B, b_C)$. Bitstrings are ordered $c[2]c[1]c[0]$ = Charlie, Bob, Alice.

3.3.3 Analysis

The results confirm the protocol's logical correctness across all four basis combinations. In each circuit, probability mass is concentrated on the four bitstrings consistent with the GHZ correlation constraint for the corresponding measurement setting, with only a small fraction of outcomes falling outside this set. The mean reconstruction fidelity of 94.0% across all circuits indicates that the deterministic product relation is satisfied in the overwhelming majority of sampled runs.

Minor deviations from the ideal distribution arise from finite sampling (shot noise) rather than any physical or numerical error. While the statevector simulator computes exact outcome probabilities, the reported frequencies are obtained from a finite number of shots (1024 in this case), leading to statistical fluctuations around the theoretical values. As a result, even outcomes that have zero probability in the ideal distribution may appear with small empirical frequency.

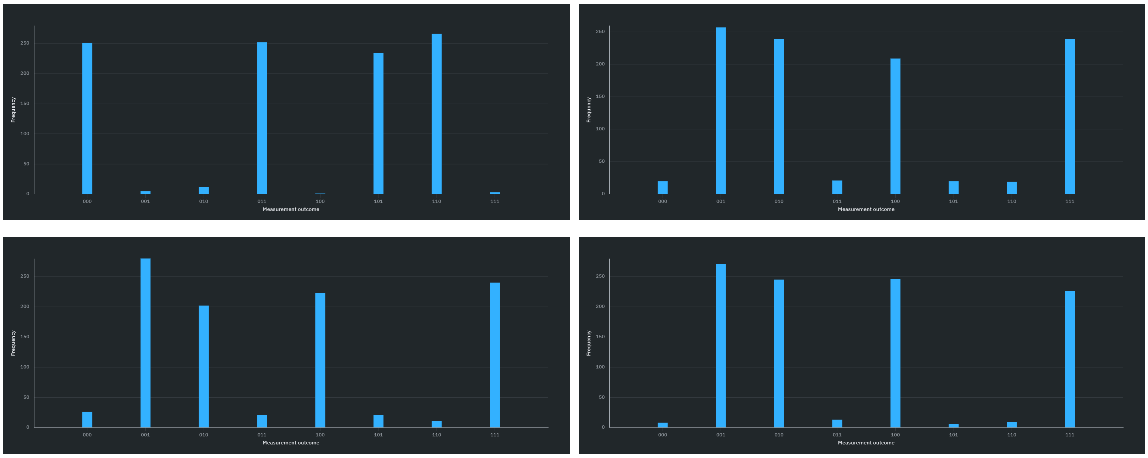

Figure 1: Measurement outcome histograms for the four HBB protocol circuits executed on the IBM Quantum Composer statevector simulator (1024 shots each). In each panel, the four dominant bars correspond to the valid bitstrings satisfying the GHZ product constraint for that basis combination. Residual counts on invalid bitstrings are attributable to floating-point precision in the statevector simulation.

Two structural features of the results are worth noting. First, within each circuit, the four valid bitstrings appear with approximately equal frequency, each accounting for roughly one quarter of total shots. This is consistent with the theoretical prediction that, conditioned on a valid basis combination, all outcomes satisfying the GHZ constraint are equiprobable. This uniformity ensures that, from the perspective of any individual party, measurement outcomes remain statistically indistinguishable and do not reveal information about the encoded secret.

Second, the XXX circuit exhibits the highest observed fidelity, while the mixed-basis circuits (involving Y measurements) show slightly lower values. In an ideal simulator, all four configurations are theoretically equivalent; the observed variation is therefore attributable to finite sampling rather than any systematic asymmetry between measurement bases.

It is important to emphasize that the statevector simulator models an idealized, noiseless execution environment, with no gate errors, decoherence, or measurement noise. The results therefore establish only that the protocol's logical structure is internally consistent and that the reconstruction rule operates as expected under ideal conditions. On physical quantum hardware, additional sources of error would reduce observed fidelities, necessitating error mitigation or fault-tolerant encoding for reliable deployment.

4 Security Analysis

The HBB protocol makes a precise security claim: a $(2, 3)$-threshold scheme in which any single party gains no information about the secret, and any external eavesdropper cannot intercept information without producing a detectable disturbance. This section establishes both guarantees formally. The argument proceeds in three parts: security against Bob acting alone, security against Charlie acting alone, and security against an external eavesdropper intercepting the quantum channel.

4.1 Security Against a Single Agent

4.1.1 Bob Cannot Determine the Secret Alone

Suppose, for contradiction, that Bob can determine s from his available information alone. Bob's available information consists of: his qubit B and its measurement outcome r_B, the publicly announced basis choices b_A and b_B, and Alice's publicly announced encoded result $\tilde{r}_A = (-1)^s r_A$.

From the reconstruction equation (32), recovering s requires evaluating:

$$(-1)^s = \frac{\tilde{r}_A \cdot r_B \cdot r_C}{c(b_A, b_B, b_C)}. \tag{33}$$

Every quantity on the right-hand side is known to Bob except r_C. For Bob to determine s, he must determine r_C.

From the GHZ correlation structure established in Section 2, the marginal distribution of r_C conditioned on any fixed values of r_A and r_B is uniform:

$$P(r_C = +1 \mid r_A, r_B) = P(r_C = -1 \mid r_A, r_B) = \frac{1}{2}. \tag{34}$$

This follows from the reduced density matrix of the BC subsystem after Alice's measurement. Tracing out Alice's qubit from the post-measurement state yields:

$$\rho_{BC} = \mathrm{Tr}_A\big(|\mathrm{GHZ}\rangle\langle\mathrm{GHZ}|\big) = \frac{1}{2}\big(|00\rangle\langle 00| + |11\rangle\langle 11|\big), \tag{35}$$

which assigns equal probability to both values of r_C regardless of r_B. Bob's estimate of $(-1)^s$ is therefore equally likely to be $+1$ or -1, which is no better than a random guess. This contradicts the assumption that Bob can determine s, and establishes that Bob alone gains zero information about the secret.

4.1.2 Charlie Cannot Determine the Secret Alone

The argument for Charlie is identical by symmetry. Charlie's available information consists of his qubit C, its measurement outcome r_C, the publicly announced basis choices b_A and b_B, and Alice's announced $\tilde{r}_A$. Recovering s from (32) requires knowing r_B, which Charlie does not have. By the same marginal uniformity argument:

$$P(r_B = +1 \mid r_A, r_C) = P(r_B = -1 \mid r_A, r_C) = \frac{1}{2}, \tag{36}$$

which follows from the reduced density matrix:

$$\rho_{BC} = \frac{1}{2}\big(|00\rangle\langle 00| + |11\rangle\langle 11|\big). \tag{37}$$

Charlie's estimate of $(-1)^s$ is equally likely to be $+1$ or -1. By contradiction, Charlie alone gains zero information about the secret.

4.2 Security Against an External Eavesdropper

Suppose there is an external eavesdropper, Eve, who intercepts the quantum channel during GHZ state distribution and attempts to extract information about the secret without being detected.

4.2.1 The No-Cloning Constraint

Eve's most natural strategy is to intercept one of the qubits in transit—say, Charlie's qubit C—measure it, and forward a substitute qubit to Charlie. The goal is to obtain a copy of r_C without disturbing the protocol.

This strategy fails for a fundamental reason: the no-cloning theorem [23] prohibits the creation of an identical copy of an unknown quantum state. Eve cannot copy qubit C before forwarding it, because the state of C is entangled with A and B and is not in a known pure state. Any attempt to clone or measure C necessarily disturbs the entangled state.

4.2.2 Measurement Disturbance

Suppose Eve intercepts qubit C and measures it in some basis before forwarding a substitute to Charlie. Eve's measurement collapses the three-qubit GHZ state. The post-measurement state of the system is no longer $|\mathrm{GHZ}\rangle$ but a product state:

$$|\mathrm{GHZ}\rangle \xrightarrow{\text{Eve measures } C} \frac{1}{\sqrt{2}}\big(|00\rangle_{AB} + |11\rangle_{AB}\big) \otimes |e\rangle_C, \tag{38}$$

where $|e\rangle_C$ is the post-measurement state of C determined by Eve's measurement outcome. The entanglement between C and the AB subsystem is destroyed.

When Charlie subsequently measures his substitute qubit, his outcome r_C is no longer correlated with r_A and r_B through the GHZ constraint. The deterministic product relation:

$$r_A \cdot r_B \cdot r_C = c(b_A, b_B, b_C) \tag{39}$$

no longer holds with certainty. Instead, Bob and Charlie's reconstruction will fail on a detectable fraction of rounds, producing outcomes inconsistent with the expected constraint. Alice, Bob, and Charlie can detect Eve's presence by publicly comparing a subset of their outcomes and checking whether the product constraint is satisfied [10].

4.2.3 Eve's Information Gain

Even setting aside detectability, Eve's measurement of qubit C yields no useful information about the secret s. Alice has not yet encoded s at the time the qubits are distributed—the secret is encoded in Step 6 of the protocol, after the quantum channel has been used. Eve's interception occurs during distribution, before s enters the protocol. Any information Eve extracts from qubit C is therefore independent of s by construction.

More precisely, qubit C prior to Alice's encoding carries only the marginal state:

$$\rho_C = \mathrm{Tr}_{AB}\big(|\mathrm{GHZ}\rangle\langle\mathrm{GHZ}|\big) = \frac{I}{2}, \tag{40}$$

which is the maximally mixed state and is independent of any information about s. Eve's measurement of this state yields a uniformly random outcome that carries zero mutual information with the secret.

4.3 Summary of Security Guarantees

The three arguments above establish the following:

- *Single-party security:* Neither Bob nor Charlie, acting alone with full knowledge of all public announcements, can determine s with probability better than $\frac{1}{2}$.

- *Eavesdropper security:* An external adversary intercepting the quantum channel during state distribution gains zero information about s, and any interception strategy that disturbs the quantum state is detectable through constraint verification.
- *Physical basis:* Both guarantees derive from properties of quantum mechanics—the marginal uniformity of GHZ correlations and the no-cloning theorem—rather than from any computational assumption. They hold against an adversary with unlimited computational power.

It is important to note the scope of these guarantees. The protocol assumes that the GHZ state is distributed faithfully, that classical communication channels are authenticated, and that participants follow the protocol honestly except for the specific deviations considered above. A fully device-independent security proof, which makes no assumptions about the internal workings of the measurement devices, requires additional machinery beyond the scope of this treatment [1].

5 Conclusion and Outlook

5.1 Summary

This paper has developed the case for Quantum Secret Sharing (QSS) as a physically grounded alternative to classical threshold schemes in high-value cryptographic infrastructure. The argument proceeded in four stages.

First, the classical problem was established. Shamir's Secret Sharing provides an information-theoretic guarantee below the reconstruction threshold, but this guarantee is narrowly scoped: it applies only to the shares as mathematical objects. It does not extend to the channels used to distribute those shares, the authentication mechanisms coordinating reconstruction, or the cryptographic primitives applied to the recovered secret. Shor's algorithm targets precisely this surrounding infrastructure. As a result, the operational security of classical threshold systems degrades even when their algebraic core remains intact.

Second, the quantum resource was introduced. GHZ states encode information in a form that is irreducibly multipartite: no two-party subset retains quantum coherence, the marginal distribution of any strict subset is uniform, and the resulting correlations cannot be reproduced by any classical shared randomness. The GHSZ no-go result strengthens this claim: the correlations required by the protocol are not merely computationally difficult to simulate, but logically incompatible with any classical model of local pre-existing values. This is what elevates the security guarantee from computational to physical.

Third, the HBB protocol was constructed and validated. The protocol operationalizes GHZ correlations into a (2,3)-threshold scheme in which a secret is reconstructed from jointly constrained measurement outcomes. Experimental verification on the IBM Quantum Composer statevector simulator confirmed logical correctness across all measurement settings, with a mean reconstruction fidelity of 94.0% over 1024 shots per circuit. While this validation is limited to simulation, it demonstrates that the protocol's reconstruction rules are internally consistent and robust to sampling noise.

Fourth, the security guarantees were formalized. Any strict subset of participants observes outcomes that are statistically independent of the secret, leaving the reconstruction equation underdetermined. An external adversary interacting with the quantum channel gains no information without introducing detectable disturbances in the correlation structure. These guarantees derive from the structure of quantum measurement and entanglement, and do not rely on assumptions about computational hardness.

Taken together, we can now proceed to discuss the potential application of QSS in financial institutions and some obstacles we are facing.

5.2 Outlook

5.2.1 Financial Applications: Interbank Settlement

The most immediate and natural deployment context for QSS is the protection of master cryptographic keys in interbank settlement infrastructure—the setting introduced in Section 1.5. In SWIFT deployments, the operational architecture already mirrors the threshold structure that QSS formalizes: multiple custodians, physical presence requirements, and a quorum-based reconstruction protocol. The classical vulnerability lies not in this architecture itself but in the channels over which shares are transmitted and the mechanisms by which custodians are authenticated. QSS addresses this vulnerability.

A concrete deployment scenario proceeds as follows. Prior to an HSM initialization ceremony, a trusted source distributes a three-qubit GHZ state among three custodians, each located at a geographically separated institution. The secret is encoded by the dealer using the HBB protocol, and each custodian retains one qubit. Reconstruction requires the physical cooperation of at least two custodians, who must perform coordinated measurements and exchange classical outcomes over authenticated channels. An adversary who compromises a single custodian's qubit gains nothing: the marginal state of any single qubit is maximally mixed and carries zero information about the master key.

This architecture is compatible with existing SWIFT operational procedures. The quantum layer governs only the share distribution phase, which currently relies on classical encrypted channels. The physical presence verification, HSM boundary enforcement proceeds unchanged. In the near term, a hybrid deployment in which QSS protects share distribution while post-quantum cryptography secures the surrounding communication infrastructure offers a meaningful security upgrade without requiring a whole replacement of existing systems [22].

Scaling this architecture to the (k, n) threshold schemes used in practice—where n may be five or more custodians and k a majority quorum—requires extensions beyond the $(2, 3)$ HBB protocol, specifically the graph state constructions and quantum error correcting code based schemes surveyed by Cleve, Gottesman, and Lo [6]. These generalizations are theoretically well established, and their practical realization represents one of the clearer near-term targets for applied quantum cryptography in the financial sector.

5.2.2 Financial Applications: Regulatory Compliance and Auditability

A property of QSS that has received comparatively little attention in the literature is its natural compatibility with regulatory and audit requirements. Financial regulators impose strict controls on access to high-value cryptographic keys: every reconstruction attempt must be logged, every participating custodian must be identified, and any unauthorized access attempt must be detected and reported. Classical threshold schemes satisfy these requirements through procedural controls and audit logs maintained on classical systems, which are themselves subject to tampering.

QSS provides a provides a tamper-evident audit mechanism that operates independently of any classical logging infrastructure. As established in Section 4, any attempt by an unauthorized party to intercept a qubit during distribution necessarily disturbs the quantum state in a detectable way. This can expose unauthorized tampering immediately. An adversary cannot intercept a share and cover their tracks, because the act of measurement collapses the entangled state and produces anomalies in the constraint verification step that Alice, Bob, and Charlie can detect by comparing a subset of their outcomes publicly [10].

This property has direct implications for regulatory frameworks governing systemically important financial institutions. Under current guidance from bodies such as the Basel Committee on Banking Supervision and national regulators, institutions must demonstrate that access to master keys is controlled and that unauthorized access attempts are detectable [2]. QSS satisfies these requirements, providing a guarantee on the independence and integrity of any classical audit system. In a threat model where an adversary has compromised classical logging infrastructure—a realistic concern given the sophistication of state-level attackers—this distinction is operationally significant.

Furthermore, the measurement disturbance property provides a natural mechanism for post-hoc verification of protocol integrity. After a key reconstruction ceremony, the participating institutions can publicly verify that the constraint relations held across all measurement rounds, confirming that no interception occurred during the session. This verification is cryptographically binding in the sense that it cannot be fabricated: producing a consistent set of GHZ measurement outcomes without access to the entangled state is precluded by the GHSZ no-go result established in Section 2.

5.2.3 Authenticated Quantum Channels at Scale

The deployment scenarios described above presuppose that GHZ states can be distributed faithfully among geographically separated institutions. In practice, this requires quantum channels capable of transmitting entangled photons with sufficient fidelity over distances relevant to financial infrastructure. Current quantum technology limits the range of entanglement distribution to tens to hundreds of kilometers before fidelity degrades to operationally unacceptable levels [5]. Scaling entanglement distribution to the distances separating, for instance, SWIFT member institutions across continents requires either significant advances in quantum repeater performance or the development of satellite-based entanglement distribution, which has been demonstrated in principle, but not at the fidelity or reliability required for financial applications [24].

5.2.4 Decoherence and Fault Tolerance

Even within achievable distances, maintaining the coherence of multipartite entangled states long enough to execute the protocol is a significant engineering challenge. GHZ states are particularly fragile: because all parties must measure before decoherence destroys the correlations, the protocol places strict timing requirements on all participants. Any interaction between the qubits and their environment during distribution induces decoherence, which manifests as a reduction in reconstruction fidelity of exactly the kind observed in physical quantum hardware. Fault-tolerant implementations of QSS, in which logical qubits are encoded in quantum error-correcting codes to protect against decoherence, have been proposed theoretically [6] but have not yet been demonstrated experimentally at the qubit counts required for practical deployment.

5.2.5 Device-Independent Security

The security analysis presented in Section 4 assumes that measurement devices operate as specified. In practice, measurement devices may be imperfectly calibrated, or deliberately compromised. A device-independent security proof, which derives security guarantees from observed measurement statistics alone without assuming anything about the internal workings of the devices, would significantly strengthen the practical security argument [1]. Device-independent QSS remains an active research direction, with partial results available for restricted adversary models but no fully general construction yet established [16]. For financial institutions, device-independent guarantees would represent a qualitative improvement over the current state of the art.

5.2.6 Integration with Post-Quantum Classical Infrastructure

Even if the above challenges are resolved, QSS will not replace classical cryptographic infrastructure overnight. Financial institutions operate on decades-long technology cycles, and the near-term migration path runs through post-quantum cryptography rather than quantum protocols. The practical question is therefore how QSS and classical schemes coexist during the transition period. Hybrid architectures, in which classical post-quantum cryptography secures the communication channels while quantum protocols provide the core secret sharing guarantee, represent a natural intermediate step [22]. Such architectures inherit the computational security of post-quantum schemes for the surrounding infrastructure while preserving the physical security guarantee of QSS for the secret itself. This will be a meaningful improvement over purely classical deployments even before fully quantum networks are available.

5.3 Final Remarks

As quantum computing hardware matures and the timeline for cryptographically relevant machines shortens, the assumptions underlying classical cryptographic infrastructure will come under increasing pressure. Post-quantum cryptography offers an important near-term remedy, but it replaces one set of unproven hardness assumptions with another. Quantum Secret Sharing offers something the classical paradigm cannot: a guarantee whose validity does not depend on what an adversary can or cannot compute, but on what the laws of physics permit.

The HBB protocol, for all its simplicity, makes this guarantee concrete. It demonstrates that the transition from computational to physical security is not merely a theoretical aspiration but a constructive possibility—one that can be implemented, verified, and reasoned about rigorously. For financial infrastructure specifically, the consequences of cryptographic failure are systemic rather than local, and thus remain an area of great attention and challenge. The open problems surveyed above are significant, but they are engineering and scalability challenges rather than fundamental obstacles.

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